

Pranati Modumudi

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EDUCATION

Columbia University

Aug 2025 - May 2027

M.S. in Computer Science, MS Thesis Track in Machine Learning and Computational Neuroscience

Relevant Coursework: Artificial Intelligence, LLM-Based Generative AI, Computation and the Brain, Theoretical Neuroscience

University of California, Berkeley

Aug 2019 - May 2023

B.A. Data Science Honors and B.A. Economics | Regents and Chancellor's Scholar | Berkeley SCET

Relevant Coursework: Data Structures; Linear Algebra and Differential Equations; Probability & Probabilistic Modeling for Data Science; Data Engineering; Natural Language Processing; Game Theory

SKILLS

- ❖ **Programming & Tools:** Python, R, SQL, Bash, Java, JavaScript, Git, Databricks, Snowflake, Tableau
 - ❖ **ML/DL:** PyTorch, TensorFlow/Keras, JAX, Scikit-learn, HuggingFace, AllenNLP, SpaCy, Neural Networks
 - ❖ **LLMs & Generative AI:** Large Language Models, Prompt Engineering, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), RLHF, Model Fine-Tuning
 - ❖ **Data & Research:** Experimental Design, Model Evaluation & Interpretation, Exploratory Data Analysis, Data Wrangling, ETL Pipelines, APIs, Big Data Systems
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INDUSTRY EXPERIENCE

Disney Streaming, New York

Data Scientist I

Aug 2023 - May 2025

- ❖ Built end-to-end machine learning pipelines for time-series and user-level causal inference using Bayesian structural models and Regression-based time-series models with 80-90% confidence intervals.
- ❖ Applied these causal inference models to various business and marketing initiatives, such as price increases and mail campaigns, contributing to higher decision efficiency by over 75%.
- ❖ Packaged reusable Python functions via PyPi and Jenkins, improving cross-team adoption and reproducibility.

OctoML, Remote

Data Science Intern

Oct 2022 - Dec 2022

- ❖ Optimized TVM machine learning compilation software by running ~100 experiments on different-sized convolution models, tuning hyperparameters to achieve ~10% $\pm$ change in convergence time.

Machine Learning Systems Intern

Mar 2022 - Jun 2022

- ❖ Diagnosed the causes of performance variance generated by auto-tuning models on machine learning compilation software TVM, as part of the Automation team.
 - ❖ Published [paper](#), which was accepted to the MLArchSys International Symposium on Computer Architecture 2022.
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RESEARCH/PROJECTS

Human Decision Modeling | Laboratory for Intelligent Imaging and Neural Computing, Columbia University

- ❖ Investigating how human physiological patterns (EEG signals and eye patterns) during risky and ambiguous decisions can inform artificial intelligence agents for prediction, reward shaping, and uncertainty estimation.

Multi-Drive Curiosity-Based RL Agent | Independent Research Project

- ❖ Self-directed research project developing biologically inspired reinforcement learning agents that integrate competing reward systems (curiosity, survival, safety) using OpenAI Gym to address AI alignment challenges, such as reward hacking.

Perceptrons and Neural Representations | Neuromatch Academy NeuroAI Project

- ❖ Comparative analysis of spiking neural networks vs CNNs in visual working memory tasks, examining performance robustness and biological plausibility using MNIST datasets.

NeMoS | Center for Computational Neuroscience Flatiron Institute

- ❖ Integrating state space models into NeMoS, a statistical modeling framework optimized for systems neuroscience, striving to make computational neuroscience more accessible.

Undergraduate Senior Honors Thesis | Redwood Center for Theoretical Neuroscience UC Berkeley

- ❖ Selected for 15-student cohort; extended hippocampal place cell modeling from rats to bats using complex-valued neural networks to capture spatial coding differences across species under Prof. Friedrich Sommer and Dr. Gautam Agarwal.